Name : Yaseen Hamdy

1-create two workspaces dev and prod

```
yaseen@192:~/iti-track/Terraform/lab2$ terraform workspace new dev
│
│ Warning: Deprecated Parameter
│
│ The parameter "dynamodb_table" is deprecated. Use parameter "use_lockfile" instead.
│

Created and switched to workspace "dev"!

You're now on a new, empty workspace. Workspaces isolate their state,
so if you run "terraform plan" Terraform will not see any existing state
for this configuration.
yaseen@192:~/iti-track/Terraform/lab2$ terraform workspace list
│
│ Warning: Deprecated Parameter
│
│ The parameter "dynamodb_table" is deprecated. Use parameter "use_lockfile" instead.
│

  default
* dev

yaseen@192:~/iti-track/Terraform/lab2$ terraform workspace new prod
│
│ Warning: Deprecated Parameter
│
│ The parameter "dynamodb_table" is deprecated. Use parameter "use_lockfile" instead.
│

Created and switched to workspace "prod"!

You're now on a new, empty workspace. Workspaces isolate their state,
so if you run "terraform plan" Terraform will not see any existing state
for this configuration.
yaseen@192:~/iti-track/Terraform/lab2$ terraform workspace list
│
│ Warning: Deprecated Parameter
│
│ The parameter "dynamodb_table" is deprecated. Use parameter "use_lockfile" instead.
│

  default
  dev
* prod
```

2-create two variable definition files(.tfvars) for the two environments

dev.tfvars > aws_region

```
aws_region = "eu-central-1"
```

prod.tfvars > aws_region

```
aws_region = "us-east-2"
```

variables.tf > variable "aws_region"

```
variable "aws_region" {
    type = string
    default = "us-east-1"
}
```

3-separate network resources into network module

main.tf > module "network"

```
module "network" {
    source = "./network/"

}
```

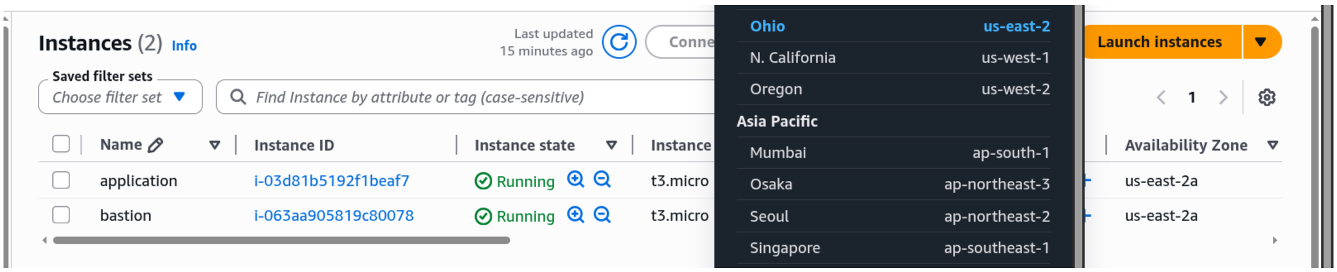
outputs of module

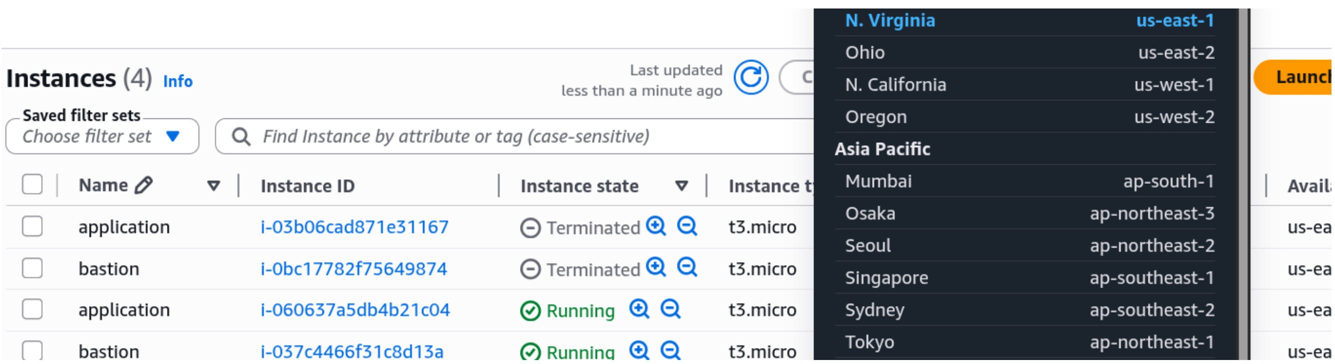
network > outputs.tf > output "public_1_subnet"

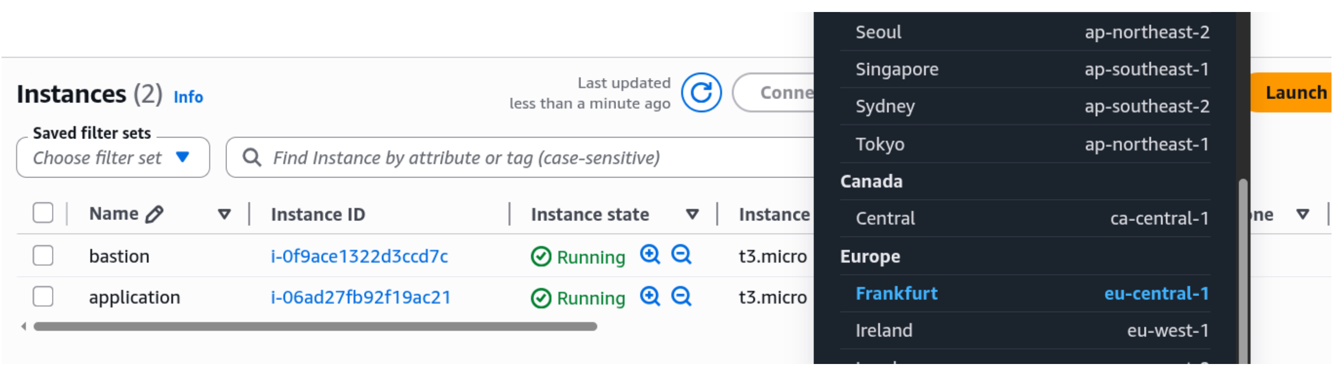
```
output "vpc" {
  value = aws_vpc.main
}
output "public_1_subnet" {

  value = aws_subnet.public_1
}

output "public_2_subnet" {

  value = aws_subnet.public_2
}

output "private_1_subnet" {

  value = aws_subnet.private_1
}

output "private_2_subnet" {

  value = aws_subnet.private_2
}

```

4-apply your code to create two environments one in us-east-1 and eu-central-1

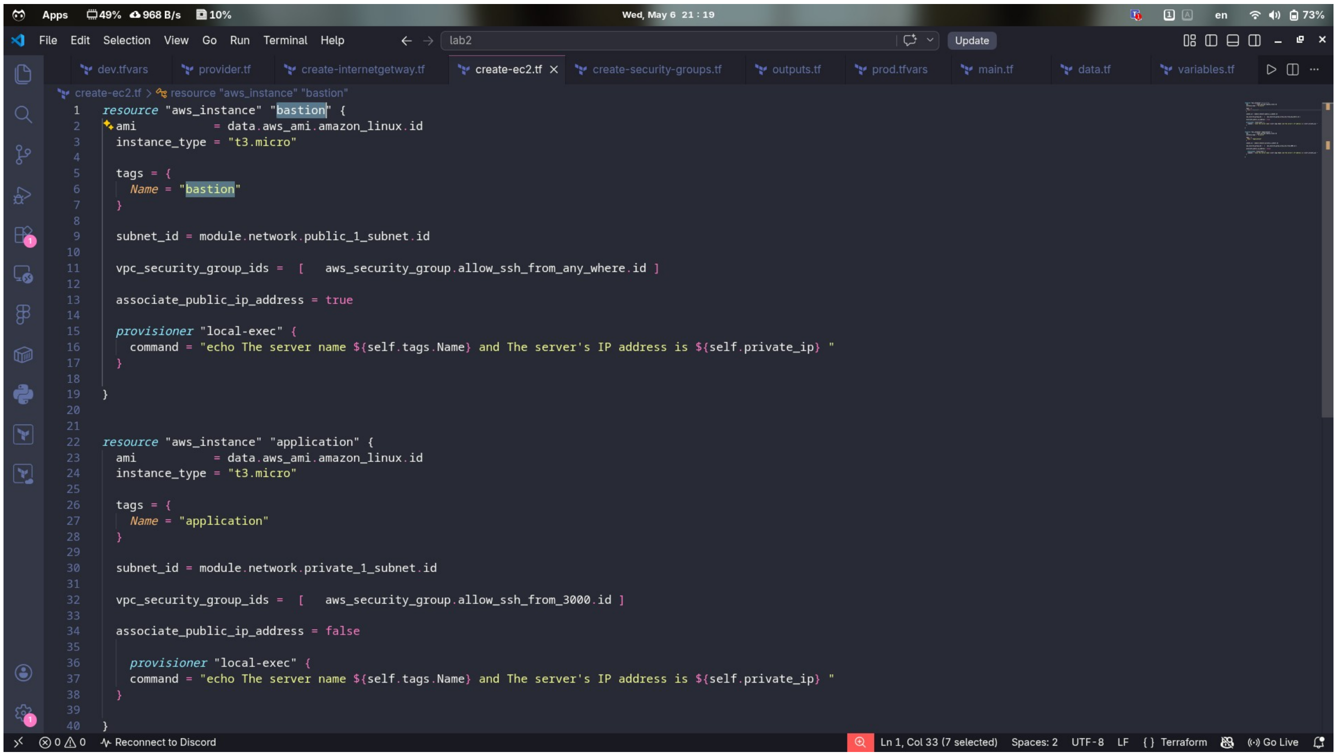
make resources in 3 regions
→ default → us-east-1
→ dev → eu-central-1
→ prod → us-east-2



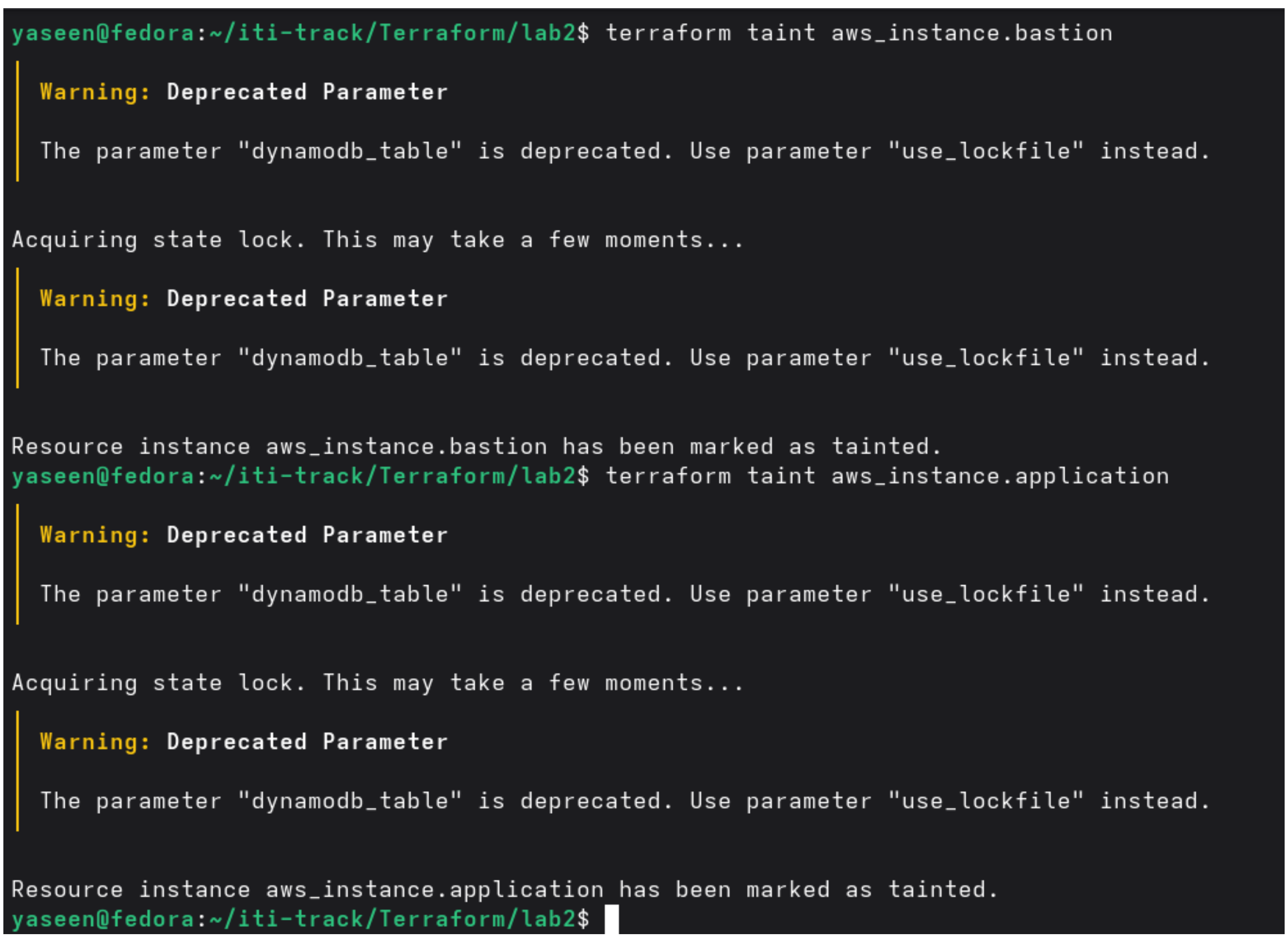




5-run local-exec provisioner to print the public_ip of bastion ec2



```
resource "aws_instance" "bastion" {
  ami           = data.aws_ami.amazon_linux.id
  instance_type = "t3.micro"

  tags = {
    Name = "bastion"
  }

  subnet_id = module.network.public_1_subnet.id

  vpc_security_group_ids =  [   aws_security_group.allow_ssh_from_any_where.id ]

  associate_public_ip_address = true

  provisioner "local-exec" {
    command = "echo The server name ${self.tags.Name} and The server's IP address is ${self.private_ip} "
  }

}


resource "aws_instance" "application" {
  ami           = data.aws_ami.amazon_linux.id
  instance_type = "t3.micro"

  tags = {
    Name = "application"
  }

  subnet_id = module.network.private_1_subnet.id

  vpc_security_group_ids =  [   aws_security_group.allow_ssh_from_3000.id ]

  associate_public_ip_address = false

    provisioner "local-exec" {
    command = "echo The server name ${self.tags.Name} and The server's IP address is ${self.private_ip} "
  }

}
```

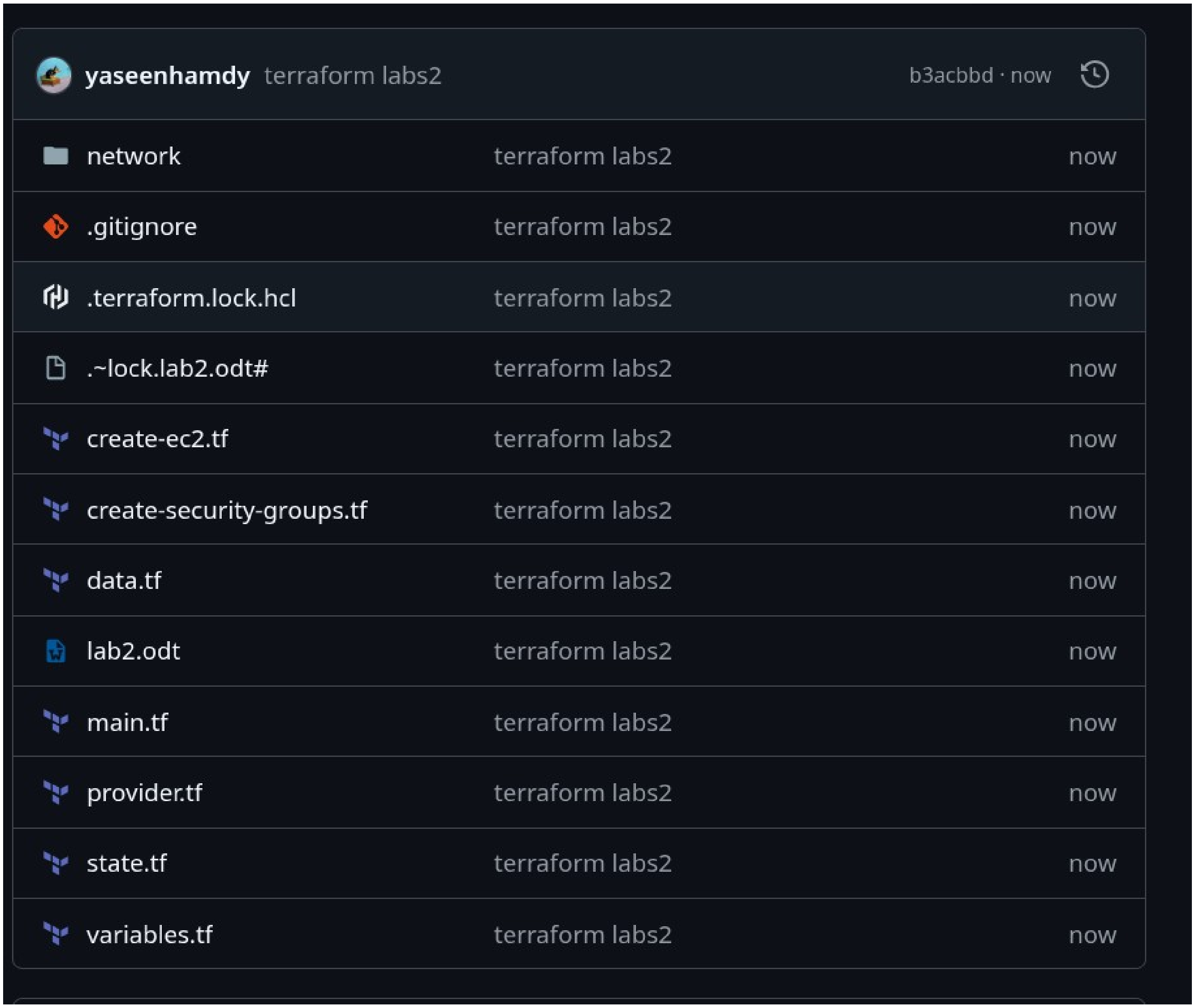


```
yaseen@fedora:~/iti-track/Terraform/lab2$ terraform taint aws_instance.bastion

 Warning: Deprecated Parameter

 The parameter "dynamodb_table" is deprecated. Use parameter "use_lockfile" instead.


Acquiring state lock. This may take a few moments...

 Warning: Deprecated Parameter

 The parameter "dynamodb_table" is deprecated. Use parameter "use_lockfile" instead.


Resource instance aws_instance.bastion has been marked as tainted.
yaseen@fedora:~/iti-track/Terraform/lab2$ terraform taint aws_instance.application

 Warning: Deprecated Parameter

 The parameter "dynamodb_table" is deprecated. Use parameter "use_lockfile" instead.


Acquiring state lock. This may take a few moments...

 Warning: Deprecated Parameter

 The parameter "dynamodb_table" is deprecated. Use parameter "use_lockfile" instead.


Resource instance aws_instance.application has been marked as tainted.
yaseen@fedora:~/iti-track/Terraform/lab2$
```

```
aws_instance.application: Provisioning with 'local-exec'...
aws_instance.application (local-exec): Executing: ["/bin/sh" "-c" "echo The server name application and The server IP address is 10.0.3.83 "]
aws_instance.application (local-exec): The server name application and The server IP address is 10.0.3.83
aws_instance.application: Creation complete after 15s [id=i-09241ffc146d27d41]
aws_instance.bastion: Still creating... [00m10s elapsed]
aws_instance.bastion: Provisioning with 'local-exec'...
aws_instance.bastion (local-exec): Executing: ["/bin/sh" "-c" "echo The server name bastion and The server IP address is 10.0.1.120 "]
aws_instance.bastion (local-exec): The server name bastion and The server IP address is 10.0.1.120
aws_instance.bastion: Creation complete after 14s [id=i-0ab00bf63fa25ffe0]
```

6- upload infrastructure code on github project

repo link : https://github.com/yaseenhamdy/terraform-labs.git



7-verify your email in ses service

create-sees.tf > ...

```
resource "aws_ses_email_identity" "email_1" {
  email = "yaseenhamdy74@gmail.com"
}

resource "aws_ses_email_identity" "email_2" {
  email = "yaseenhamdy888@gmail.com"
}
```

8- create lambda function to send email

→ IAM policy

modules > lambda > IAM_policy.tf > resource "aws_iam_policy" "lambda_ses_policy

```
resource "aws_iam_policy" "lambda_ses_policy" {
  name        = "lambda_ses_policy"
  path        = "/"
  description = "IAM policy for send emails from Lambda"

  policy = jsonencode({
    Version = "2012-10-17"
    Statement = [
      {
        Effect = "Allow"
        Action = [
          "ses:SendEmail",
          "ses:SendRawEmail"
        ]
        Resource = "*"
      }
    ]
  })
}
```

→ IAM Role

modules > lambda > IAM_role.tf > ...

```
resource "aws_iam_role" "lambda_role" {
  name = "lambda_role"

  assume_role_policy = jsonencode({
    Version = "2012-10-17"
    Statement = [
      {
        Action = "sts:AssumeRole"
        Effect = "Allow"
        Principal = {
          Service = "lambda.amazonaws.com"
        }
      }
    ]
  })
}
```

→ attach policy in role

modules > lambda > role_policy_attach.tf > ...

```
resource "aws_iam_role_policy_attachment" "lambda_role_policy" {
  role       = aws_iam_role.lambda_role.name
  policy_arn = aws_iam_policy.lambda_ses_policy.arn
}
```

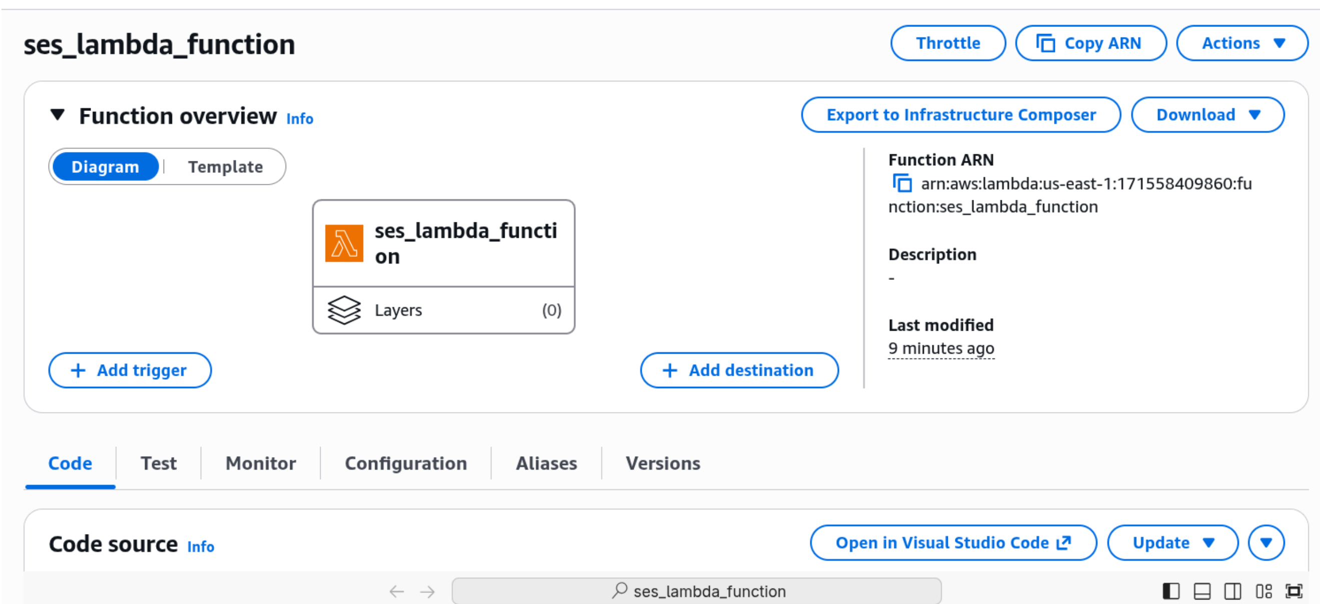
→ archiving code file

modules > lambda > data-lambda-archiving.tf > data "archive_file" "lambda_archiving"

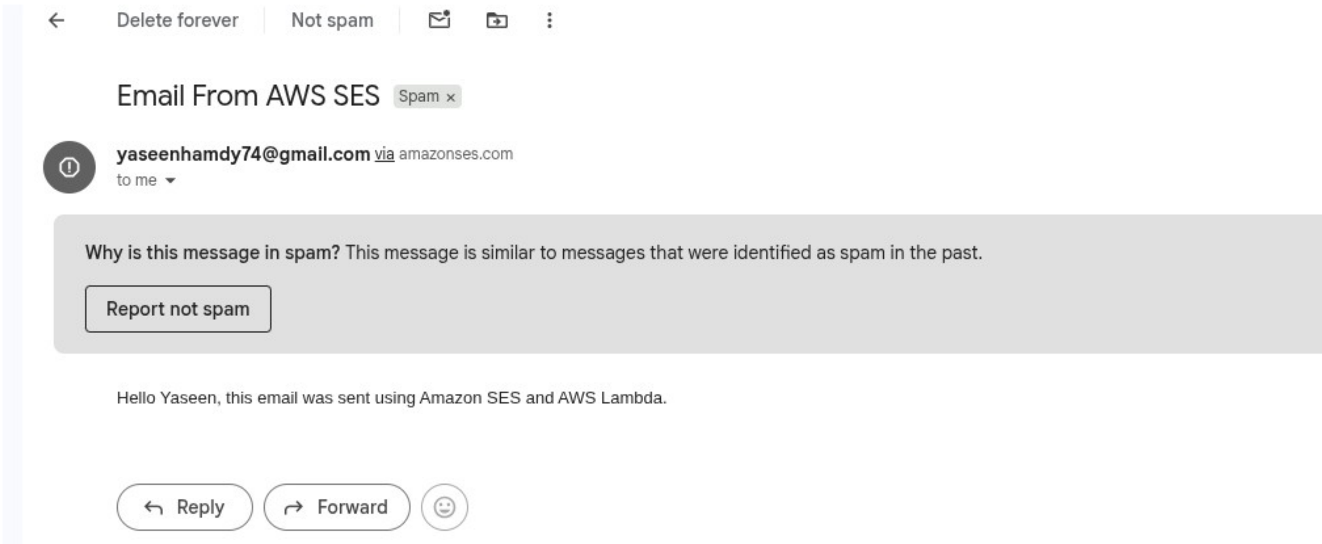
```
data "archive_file" "lambda_archiving" {
  type        = "zip"
  source_file = "${path.module}/send_email.py"
  output_path = "${path.module}/send_email.zip"
}
```

→ create lambda function

```
modules > lambda > lambda-func.tf > resource "aws_lambda_function" "example"
resource "aws_lambda_function" "example" {
  filename      = data.archive_file.lambda_archiving.output_path
  function_name = "ses_lambda_function"
  role          = aws_iam_role.lambda_role.arn
  handler       = "send_email.lambda_handler"
  code_sha256   = data.archive_file.lambda_archiving.output_base64sha256

  runtime = "python3.12"

}
```



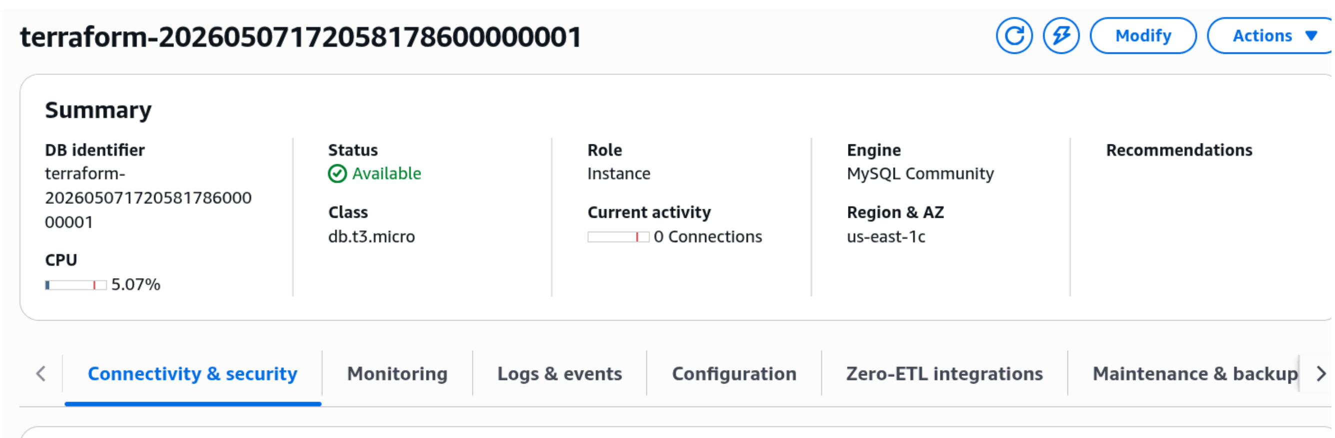
9-create trigger to detect changes in state file and send the email



10-create rds (mysql)

modules > rds > create-sql-rds.tf > resource "aws_db_instance" "sqlDB"

```
resource "aws_db_instance" "sqlDB" {
  allocated_storage    = 10
  db_name              = var.db_name
  engine               = "mysql"
  engine_version       = "8.0"
  instance_class       = "db.t3.micro"
  username             = var.db_username
  password             = var.db_password
  parameter_group_name = "default.mysql8.0"
  skip_final_snapshot  = true
}
```



11-create elastic cache (redis)

modules > rds > create-sql-rds.tf > ...

```
resource "aws_db_instance" "sqlDB" {
  allocated_storage    = 10
  db_name              = var.db_name
  engine               = "mysql"
  engine_version       = "8.0"
  instance_class       = "db.t3.micro"
  username             = var.db_username
  password             = var.db_password
  parameter_group_name = "default.mysql8.0"
  skip_final_snapshot  = true
}
```

app-redis Info

Upgrade to Valkey | Modify | Act

▼ Cluster details

Cluster name app-redis	**Description** -	**Node type** cache.m4.large	**Status** Available
Engine Redis	**Engine version** 7.1.0	**Global datastore** -	**Global datastore role** -
Update status Up to date	**Cluster mode** Disabled	**Shards** 0	**Number of nodes** 1
Data tiering Disabled	**Multi-AZ** Disabled	**Auto-failover** Disabled	**Encryption in transit** Disabled
Encryption at rest Disabled	**Parameter group** default.redis7	**Outpost ARN** -	**Configuration endpoint** -
Primary endpoint -	**Reader endpoint** -	**ARN** arn:aws:elasticache:us-east-1:171558409860:cluster:app-redis	**Data migration** No active migrations